

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR-A&APEX, ALL IITS **DT: 04/01/2026**
Time: 3 Hrs **GTM – 08 (TBT-1)** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A parallel plate air capacitor is connected to a battery. The quantities charge, capacitance, electric field and energy associated with this capacitor are given by Q_0, C_0, E_0 and U_0 respectively. A dielectric slab is now introduced to fill the space between the plates with battery still in connection. The corresponding quantities now given by Q, C, E and U are related to the previous one as

(A) $C > C_0$; $Q > Q_0$; $E > E_0$; $U > U_0$

(B) $C > C_0$; $Q > Q_0$; $E = E_0$; $U > U_0$

(C) $C > C_0$; $Q > Q_0$; $E < E_0$; $U > U_0$

(D) $C > C_0$; $Q < Q_0$; $E = E_0$; $U < U_0$

2. A ray of light travelling in water is incident on its surface open to air. The angle of incidence is θ, which is less than the critical angle, then there will be

(A) only a reflected ray and no refracted ray

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(B) only a refracted ray and no reflected ray

(C) a reflected ray and refracted ray and the angle between them would be less than $180°-2\theta$

(D) a reflected ray and a refracted ray and the angle between them would be greater than $180°-2\theta$

3. Consider two identical circular loops of same radius, with the second (right side) coil rotated slightly clockwise relative to the first when looked from above as shown in figure (a). A large current is suddenly injected into the left side loop. What happens to the right side loop?

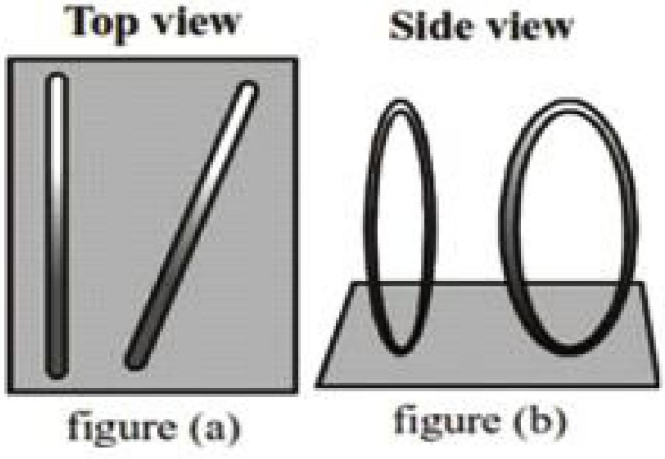


figure (a) figure (b)

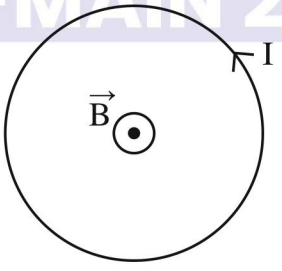
A) Force to the left, torque rotates clockwise (in top view)

B) Force to the right, torque rotates clockwise (in top view)

C) Force to the right, torque rotates counterclockwise (in top view)

D) Can't tell without knowing which direction current injected into left loop

4. A current carrying circular coil is placed on a smooth horizontal table in a uniform magnetic field as shown in the figure. The direction of magnetic field is out of the plane of loop.



The coil has a tendency to

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(A) contract (B) expand

(C) neither contract nor expand (D) translate towards right

5. A point source of light is used in a photoelectric effect. If the source is removed farther from the emitting metal, the stopping potential

(A) will increase (B) will decrease

(C) will remain constant (D) will either increase or decrease

6. A magnetic dipole in a constant magnetic field has

(A) maximum potential energy when the torque is maximum

(B) zero potential energy when the torque is minimum

(C) zero potential energy when the torque is maximum

(D) minimum potential energy when the torque is maximum

7. STATEMENT-1: When a coil is connected to a cell, no current flows through it initially.

STATEMENT-2: When a coil is connected to a cell, the initial emf induced in it is equal to the emf of the cell.

A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

B) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1

C) Statement-1 is True, Statement-2 is False

D) Statement-1 is False, Statement-2 is True

8. An electron makes a transition from an excited state to the ground state of a hydrogen-like atom/ion

(A) Kinetic energy decreases, potential energy increases but total energy remains same

(B) Kinetic energy and total energy decrease but potential energy increases

(C) Kinetic energy increase but potential energy and total energy decrease

(D) Kinetic energy, potential energy and total energy decrease

9. Light emerges out of a convex lens when a point source of light kept at its focus. The shape of wavefront of the light is

(A) Both spherical and cylindrical (B) Cylindrical

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(C) Spherical (D) Plane

10. STATEMENT-1: It is not possible for a charged particle to move in a circular path around a long straight conductor carrying current.

STATEMENT-2: The electromagnetic force on a moving particle is normal to its plane of rotation.

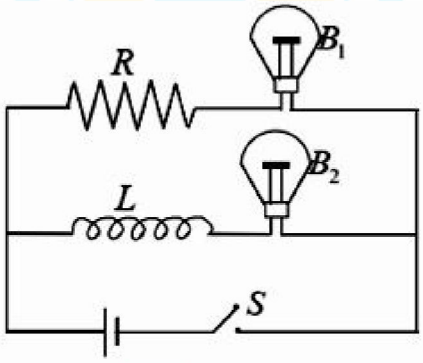
A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

B) Statement-1 is True, Statement-2 is True; Statement-2 is not a correct explanation for Statement-1

C) Statement-1 is True, Statement-2 is False

D) Statement-1 is False, Statement-2 is True

11. In the given circuit R is a resistor, L is an inductor and B_1, B_2 are two bulbs. If the switch S is closed for a long time and then opened



A) Both B_1 and B_2 die out promptly

B) Both B_1 and B_2 die out with some delay

C) B_1 dies out promptly but B_2 with some delay

D) B_2 dies out promptly but B_1 with some delay

12. Electric and magnetic field of an electromagnetic wave are

A) in phase and perpendicular to each other

B) in phase and parallel to each other

C) in opposite phase and parallel to each other

D) in opposite phase and perpendicular to each other

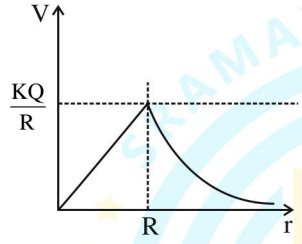
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13. The wavelength of a matter wave is independent of

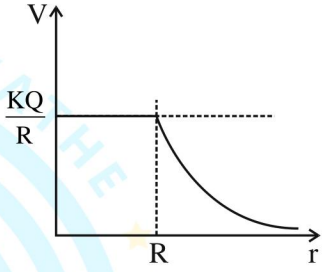
A) Mass B) Speed C) Momentum D) charge

14. Charge Q is uniformly distributed throughout the volume of a non-conducting sphere of radius R. Choose the correct graph for electrostatic potential (V) versus distance (r) from the centre of the sphere

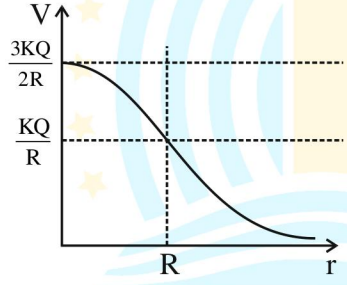
(A)



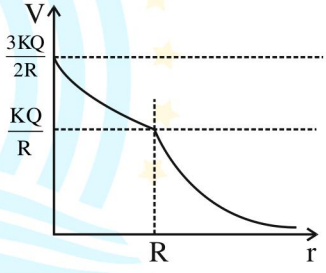
(B)



(C)



(D)



15. Two inputs of a NAND gate are shorted. The gate is equivalent to

A) NOT gate B) NOR gate C) XOR gate D) OR gate

16. Needles N_1, N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will

A) Attract equally three of them

B) Attract N_1 and N_2 strongly but repel N_3

C) Attract N_1 strongly, N_2 weakly and repel N_3 weakly

D) Attract N_1 strongly, but repel N_2 and N_3 weakly.

17. If the number of turns per unit length of a long solenoid is doubled, the self-inductance of the solenoid will

(A) remain unchanged (B) be halved

(C) be doubled (D) become four times

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18. A light wave and a sound wave have same frequency f and their wavelengths are respectively λ_L and λ_S ; then

A) $\lambda_L = \lambda_S$ B) $\lambda_L > \lambda_S$ C) $\lambda_L < \lambda_S$ D) $\lambda_L = 2\lambda_S$

19. Particles and their anti-particles have:

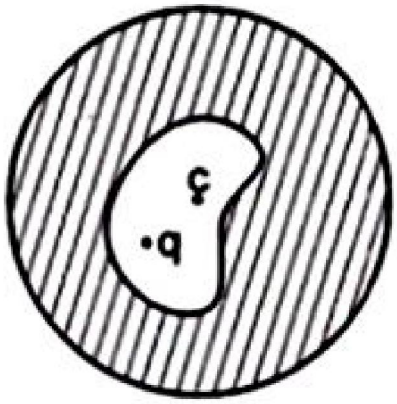
A) The same masses but opposite spins

B) The same masses but opposite magnetic moments

C) The same masses and same magnetic moments

D) Opposite spins and same magnetic moments

20. The figure shows a charge q placed inside a cavity in an uncharged conductor. Now if an external electric field is switched on



A) Only induced charge on outer surface will redistribute

B) Only induced charge on inner surface will redistribute

C) Both induced charge on outer and inner surface will redistribute

D) Force on charge q placed inside the cavity will change.

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
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Note: For Questions 21 to 25, follow these rules while marking answers in the OMR sheet:

If only option 1 is correct, bubble "1". If options 1 and 2 are correct, bubble "12". If options 1, 2, and 3 are correct, bubble "123". If options 1, 2, 3, and 4 are correct, bubble "1234". If options 1 and 3 are correct, bubble "13". If options 2 and 4 are correct, bubble "24". If options 3 and 4 are correct, bubble "34"., and so on Bubble all correct option numbers together as a single answer.

21. Choose the correct statement among the following statements

1. Emf of a cell is the potential difference between its positive and negative electrodes in an open circuit
2. Internal resistance of dry cells is much higher than common electrolytic cells
3. When a current passes from positive to negative terminal of a cell inside it, the terminal potential difference is less than its emf
4. When a current passes from negative to positive terminal of a cell inside it, the terminal potential difference is less than its emf

22. A light ray of wavelength λ is passing through a pin hole of diameter D and the effect is observed on a screen placed at a distance L from the pin hole. The approximations of geometrical optics are applicable if:

1. $D >> \lambda$ 2. $\frac{L\lambda}{D^2} = 1$ 3. $\frac{L\lambda}{D^2} << 1$ 4. $\frac{L\lambda}{D^2} >> 1$

23. An electron goes undeviated with a velocity $(\vec{V})$ in a region containing non-zero electric field $(\vec{E})$ and magnetic field $(\vec{B})$. Choose the correct option. (Neglect gravity)

1. All the three vectors $\vec{V}$, $\vec{E}$ and $\vec{B}$ must be perpendicular to each other

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2. $\vec{B}$ must be perpendicular to $\vec{V}$ and $\vec{E}$

3. $\vec{E}$ must be perpendicular to $\vec{V}$ and $\vec{B}$

4. $\vec{V}$ must be perpendicular to $\vec{E}$ and $\vec{B}$

24. Choose the correct statement

1. The electric lines of force entering into a Gaussian surface provide negative flux

2. A charge 'q' is placed at the centre of a cube. The electric flux through all the faces will be the same

3. In a uniform electric field net flux through a closed Gaussian surface containing no net charge is zero

4. When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux

25. In a series LCR circuit

1. Current may lag behind voltage

2. Current and voltage may be in phase

3. Current lags behind the voltage if $\omega > \frac{1}{\sqrt{LC}}$

4. Current lags behind the voltage if $\omega < \frac{1}{\sqrt{LC}}$

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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26. Choose the incorrect statement regarding brown ring test:

A) Freshly prepared ferrous sulphate is used to avoid aerial oxidation of Fe^{2+} into Fe^{3+}

B) Ring is formed due to nitroso ferrous sulphate complex

C) Spin only magnetic moment of the brown ring complex is $\sqrt{15}\,BM$.

D) Concentrated nitric acid is used in the test.

27. Select the correct statement (s) regarding the function of Galvanic cells

a) Salt bridge allows the transfer of charge between the two half cells and prevents accumulation of charge in any of the half cells.

b) Saturated solution of KCl is used in salt bridge because the $K^{+}_{(aq)}$ and $Cl^{-}_{(aq)}$ have equal ionic mobility.

c) Anode is negative charged and cathode is positive charged.

d) e^{-} transfer takes place from cathode to anode.

A) a, b, c and d B) b, c and d only

C) a, b and c only D) a and b only

28. Find out the incorrect statements according to collision theory:

A) The reactant molecules are assumed to be hard spheres and reaction is postulated to occur when molecules collide with each other.

B) The number of collisions per second per unit volume of the reaction mixture is known as collision frequency.

C) Arrhenius constant is related to collision frequency.

D) $k = Z_{AB}e^{-E_a/RT}$ is the equation that predicts the rate constant fairly for all types of reactions.

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29. Which of the following statement is incorrect?

A) SRP values of halogens $X_2(g)/X^-(aq)$: $F_2 > Cl_2 > Br_2 > I_2$

B) Bond dissociation enthalpy of $Cl_2 > F_2 > Br_2 > I_2$

C) Boiling points of $I_2 > Br_2 > Cl_2 > F_2$

D) Reducing power of $I^- > Br^- > Cl^- > F^-$

30. Cerium (Z = 58) is an important member of lanthanoids, which of the following statements about cerium is incorrect?

A) The common oxidation states for cerium are + 3 and +4

B) The +3 oxidation state of cerium is more stable than +4

C) The +4 oxidation state of cerium is not known in solutions

D) Cerium (IV) acts as an oxidizing agent.

31. Regarding lyophilic and lyophobic sols the correct statements are

a) stability of lyophilic sol is primarily due to the solvation of colloidal particles by the solvent

b) stability of lyophobic sol is due to presence of charge on the colloidal particles.

c) Starch and As_2S_3 sol are negatively charged whereas hydrated ferric hydroxide sol obtained by addition of excess of hot water to $FeCl_3$ is positively charged.

d) As_2S_3 sol can be prepared by passing $H_2S_{(g)}$ to a boiling aqueous solution of As_2O_3

A) All correct B) a and b only C) a, b and c only D) d only

32. The amino acid alanine has two isomers, α-alanine and β-alanine. When equal masses of these two compounds are dissolved in equal mass of a solvent, the solution of α-alanine freezes at relatively lower temperature. Which form, α-alanine or β-alanine, has the larger equilibrium constant for ionization?

(A) α-alanine (B) β-alanine

(C) same for both (D) unpredictable

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33. Which of the following compound does not give colour in phthalein dye test

A) o–Cresol B) m–Cresol C) Resorcinol D) p–Cresol

34. The complex $\left[Co\left(NH_3\right)_4\left(NO_2\right)_2\right]Cl$, exhibits

A) Ionization isomerism, geometrical isomerism and optical isomerism

B) Linkage isomerism, geometrical isomerism and optical isomerism

C) Linkage isomerism, geometrical isomerism and ionization isomerism

D) Optical isomerism, ionization isomerism and Linkage isomerism

35. Assertion (A): Aniline on nitration gives meta nitro aniline in maximum yield.

Reason (R): $-NH_3^+$ acts as meta directing group.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

36. A student performs qualitative tests on an unknown organic compound suspected to contain a carbonyl group. The following observations are considered for interpretation. Which of the following observation is most suitable to distinguish aldehyde and ketone

A) Aldehydes and ketones both give precipitates with 2,4-dinitrophenylhydrazine, but ketones react faster due to greater stability of the resulting hydrazone.

B) Aldehydes are readily oxidized by mild oxidizing agents such as Tollens' reagent, whereas most simple ketones do not undergo oxidation under these conditions.

C) Sodium nitroprusside gives a wine-red colour with aldehydes but not with ketones, making it a confirmatory test for aldehydes.

D) Fehling's solution can be used universally to distinguish aldehydes from ketones because all aldehydes give a red precipitate of Cu_2O while ketones never do.

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37. Acetone and carbon disulphide (CS_2) form a binary liquid solution. Given that the normal boiling point (T_b) of pure acetone is lower than that of pure CS_2, which of the following statements is correct?

A) The boiling point of any mixture of these two liquids will always lie between the boiling points of pure acetone and pure CS_2.

B) At the azeotropic composition, the solution exhibits the lowest vapor pressure of any composition in the system at a constant temperature.

C) Starting with a mixture containing an excess of CS_2 (relative to the azeotropic composition), fractional distillation will eventually yield pure CS_2 as the residue in the boiling flask.

D) Upon mixing pure acetone and pure CS_2 in an adiabatic container, the temperature of the resulting solution will increase.

38. Which one of the following is incorrect?

A) $\left[Fe(CN)_6\right]^{4-}$ and $\left[Fe(H_2O)_6\right]^{3+}$ have same number of unpaired electrons in central atom.

B) A solution of $\left[Ni(H_2O)_6\right]^{2+}$ is green but a solution of $\left[Ni(CN)_4\right]^{2-}$ is colourless

C) $\left[Cr(NH_3)_6\right]^{3+}$ is paramagnetic while $\left[Ni(CN)_4\right]^{2-}$ is diamagnetic

D) d – orbital occupation of the central metal ion in the complex $\left[CoF_4\right]^{2-}$ is $t_{2g}^5 e_g^2$

39. Statement-1: If the order of reaction is zero then the degree of dissociation will be independent upon initial concentration.

Statement-2: The degree of dissociation in a Zero order reaction is given by $\alpha = \frac{kt}{c_0}$

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

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(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

40. Assertion(A): The complex ion $\left[Fe(CN)_6\right]^{3-}$ shows a spin-only magnetic moment of approximately 1.73BM.

Reason(R): The cyanide ion $\left(CN^-\right)$ is a π - donor ligand, its π - donation increases electron – repulsion and reduces the octahedral splitting energy Δ_0 which causes maximum pairing in $Fe^{3+}\left(d^5\right)$.

Choose the correct option:

A) A and R are true; R is correct explanation of A

B) A and R are true; R is not correct explanation of A

C) A is true ; R is false

D) A is false ; R is true

41. Comparative analysis of d^4 systems consider the $E^o\left(M^{3+}/M^{2+}\right)$ values for Chromium and Manganese. In aqueous solution, the Cr^{2+} ion acts as a strong reducing agent, while the Mn^{3+} ion acts as a strong oxidizing agent. Which of the following is the most accurate explanation for this observation?

A) The d^3 configuration of Cr^{3+} has a higher stability due to the half-filled t_{2g} level in an octahedral field.

B) Mn^{3+} has a higher hydration enthalpy than Cr^{3+}, making it more stable in its higher oxidation state.

C) The d^5 configuration of Mn^{2+} is less stable than the d^3 configuration of Cr^{3+}.

D) The sum of the first three ionization enthalpies is lower for Manganese than for Chromium.

42. In which of the following set of cations, all of them will give precipitate either in $H_2S\,/\,dil.\ HCl$ or $H_2S\,/\,NH_4OH$

A) $Pb^{2+}, Na^+, Zn^{2+}, Ni^{2+}$ B) $Pb^{2+}, Cu^{2+}, Zn^{2+}, Ni^{2+}$

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C) $Mg^{2+}, Na^{+}, Zn^{2+}, Ni^{2+}$ D) $Fe^{3+}, Na^{+}, Zn^{2+}, NH_4^{+}$

43. Which of the following statements is **incorrect**?

A) In aqueous solution, Cu^{2+} is more stable than Cu^{+} due to its much higher hydration enthalpy

B) H_3PO_3 is a tribasic acid because it contains three hydrogen atoms.

C) Zr and Hf have nearly identical atomic radii due to lanthanoid contraction.

D) BiH_3 is the strongest reducing agent among group 15 hydrides.

44.

$$\text{Set(I)}\begin{cases}\text{(i) } VF_5 + H_2O \longrightarrow VOF_3 + 2HF \\ \text{(ii) } 2VOF_3 + 3H_2O \longrightarrow V_2O_5 + 6HF\end{cases}$$

$$\text{Set(II)}\begin{cases}\text{(iii) } CoF_3 \text{ exist due to high lattice energy} \\ \text{(iv) } CrF_6,\ VF_5,\ TiF_4 \text{ exist due to high bond enthalpy}\end{cases}$$

$$\text{Set(III)}\begin{cases}\text{(v) } VF_2 \text{ does not exist but } VCl_2,\ VBr_2,\ VI_2 \text{ exist} \\ \text{(vi) } VOCl_3 \text{ exist but } VOBr_3 \text{ does not}\end{cases}$$

The set where both reaction/fact is correct

(A) Only set(I) (B) Only Set (I) and (II)

(C) Only set (II) and set (III) (D) All set (I), set (II) and set (III)

45. Two volatile liquids A and B form an ideal solution at temperature T.

The vapour pressures of the pure liquids at this temperature satisfy: $P_B^0 > P_A^0$

At equilibrium, the vapour phase has mole fractions y_A and y_B.

Which of the following statements is always correct?

A) If $x_A = x_B$, then necessarily $y_A = y_B$.

B) For all compositions, $\frac{y_A}{y_B} > \frac{x_A}{x_B}$

C) Increasing the total pressure by adding an inert gas at constant T and volume increases y_B.

D) For all compositions, $y_B > x_B$

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SECTION – II
(Numerical Value Answer Type)

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46. Find the number of basic radicals among the following, which can form complex on adding excess of KCN

$Pb^{2+}, Ag^{+}, Fe^{2+}, Fe^{3+}, Cu^{2+}, Ni^{2+}$

47. An organic solid containing carbon and hydrogen, and possibly one or more of nitrogen (N), sulphur (S), and halogens (Cl/Br/I), is fused with metallic sodium to prepare the sodium fusion extract. The following observations are recorded:

(i) A portion of the extract gives a brown ring at the junction of two liquid layers on treatment with freshly prepared $FeSO_4$ solution followed by careful addition of concentrated H_2SO_4.

(ii) Another portion of the extract gives a black precipitate on addition of acetic acid followed by lead(II) acetate solution.

(iii) A third portion of the extract, when acidified with dilute HNO_3 and treated with $AgNO_3$, gives a white precipitate which is soluble in dilute aqueous ammonia.

Find the value of (X + Y) where:

X = Total number of extra elements (N, S, Cl, Br, I) confirmed to be present in the organic compound based on the above observations.

Y = Group number (in the qualitative inorganic analysis scheme) to which the cation of black precipitate in observation (ii) belongs.

48. An aqueous solution of 'A' is mixed with another equimolal solution of 'B'. In how many of the following cases, does the elevation in boiling point of mixture is more than the initial solution (B) [Assume $\alpha = 1$ for all the electrolytes in all solutions]

(i) $BaCl_2(A)$ mixed with $Na_2SO_4(B)$ (1:1 mole ratio)

(ii) $Al_2(SO_4)_3(A)$ mixed with $BaCl_2(B)$ (1:1 mole ratio)

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(iii) $AgNO_3(A)$ mixed with KCN (B) (2:1 mole ratio)

(iv) NaCl (A) mixed with $AgNO_3(B)$ (1:1 mole ratio)

(v) $CaCl_2(A)$ mixed with $Na_2SO_4(B)$ (1:1 mole ratio)

(vi) $AlCl_3(A)$ mixed with NaOH (B) (3:1 mole ratio)

49. A transition metal M from the 3d series has a ground-state electronic configuration such that the sum of electrons in its (n - 1)d and ns subshells is 6. This metal forms a stable aqueous cation M^{x+} which exhibits a "spin-only" magnetic moment of exactly $\sqrt{15}$ BM.

If y represents the total number of unpaired electrons present in a neutral, isolated gaseous atom of the same metal M, calculate the value of (x + y):

50. A hypothetical reaction:

$A_2 + B_2 \rightarrow 2AB$

Follows mechanism as given below:

$A_2 \underset{}{\overset{K_C}{\rightleftharpoons}} A + A$ (fast) (Kc - is equilibrium constant)

$A + B_2 \xrightarrow{k_1} AB + B$ .....(slow) (k_1- rate constant)

$A + B \xrightarrow{k_2} AB$ (fast) (k_2 - rate constant)

If the overall order of reaction is X, the value $10X$ is____

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MATHEMATICS **MAX.MARKS: 100**

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. **STATEMENT – I:** Let the solution curve $y=f(x)$ of the differential equation

$\frac{dy}{dx}+\frac{xy}{x^2-1}=\frac{15x^4+4x^3+4x+1}{\sqrt{1-x^2}}, x\in(-1,1)$ passes through the origin. Then $\int_{-1/2}^{1/2}\frac{f(x)dx}{(x^2+2)}$ is equal to $\frac{\pi}{6}-\frac{\sqrt{3}}{4}$

STATEMENT – II: Let f(x) is continuous for all x, then $\int_a^b f(x)dx$, equals to

$\int_a^c f(x)dx+\int_b^c f(x)dx$

Then which of the following is true

A) STATEMENT – I is false and STATEMENT – II is true

B) STATEMENT – I is true and STATEMENT – II is false

C) Both STATEMENT – I and STATEMENT – II are false

D) Both STATEMENT – I and STATEMENT – II are true

52. Assertion (A): For $2<a<5, \int\frac{dx}{x^2+2(a-3)x+(a-1)}=A\,ln|g(x)|+C$; where A is constant and C is integration constant, where $g(x)=\left|\frac{(x+a-3)-f(a)}{(x+a-3)+f(a)}\right|$ and $f(a)=\sqrt{a^2-7a+10}$.

Reason (R): For $2<a<5, \frac{1}{x^2+2(a-3)x+(a-1)}$ is a continuous functon.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true and R is not the correct explanation of A

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C) A is true but R is false D) A is false but R is true

53. Two players A and B toss a coin which has a probability p of showing head $(0<p<1)$, in the cyclic order A, A, B, A, A, B, …. till a head appears. Let x denotes the probability that A gets the head first and y that of B, then: (read the statements below)

I) $x+y=1$ II) $x>y$ for all p

III) $x>\frac{1}{2}$ for all p IV) $y>\frac{1}{2}$ for all p

Then number of correct statements

A) 1 B) 3 C) 4 D) 2

54. **STATEMENT – I:** Let A, B are two fixed points and PA + PB = k, then if AB < k, locus of P is an ellipse

STATEMENT – II: Let A, B are two fixed points and $|PA-PB|=k$, then if AB > k, locus of P is a hyperbola

Then which of the following is true

A) Both STATEMENT – I and STATEMENT – II are true

B) Both STATEMENT – I and STATEMENT – II are false

C) STATEMENT – I is false and STATEMENT – II is true

D) STATEMENT – I is true and STATEMENT – II is false

55. **STATEMENT – I:** If $\int\frac{x^2\left(x^6+x^5-1\right)}{\left(2x^6+3x^5+2\right)^2}dx=f(x)+C$ then

$f(x)=\frac{1}{6}\left(2x^3+3x^2+2x^{-3}\right)^{-1}$ (where C is integration constant)

STATEMENT – II: Let f and g are two functions in x. If f is differentiable and g is integrable, then $\int f\cdot g\,dx$, equals to $f\cdot\int g\,dx-\int\left(f\cdot\frac{dg}{dx}\right)dx$

Then which of the following is true

A) STATEMENT – I is false and STATEMENT – II is true

B) STATEMENT – I is true and STATEMENT – II is false

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C) Both STATEMENT – I and STATEMENT – II are false

D) Both STATEMENT – I and STATEMENT – II are true

56. **STATEMENT – I :** Let $p\left(\frac{2\sqrt{3}}{\sqrt{7}}, \frac{6}{\sqrt{7}}\right), Q, R$ and S be four points on the ellipse $9x^2 + 4y^2 = 36$. Let PQ and RS be mutually perpendicular and pass through the origin. If $\frac{1}{(PQ)^2} + \frac{1}{(RS)^2} = \frac{p}{q}$, where p and q are coprime, then $p + q = 157$

STATEMENT – II: Let a double-napped right circular hollow cone of having semi vertical angle α and extending indefinitely far in both directions. This cone is intersected by a plane (Plane is not passing through vertex of cone). If β be the angle made by the intersecting plane with the vertical axis of the cone, then the cross section is a parabola when $\alpha = \beta$

Then which of the following is true

A) Both STATEMENT – I and STATEMENT – II are true

B) Both STATEMENT – I and STATEMENT – II are false

C) STATEMENT – I is false and STATEMENT – II is true

D) STATEMENT – I is true and STATEMENT – II is false

57. Statement 1: If both roots of the quadratic equation $x^2 - 2kx + k^2 + k - 5 = 0$, are less than 5, then $k \in (-\infty, 4)$

Statement 2: If both roots of $ax^2 + bx + c = 0,\ a \neq 0,\ b, c \in R$ are less than k, then (i) $b^2 - 4ac \geq 0$, (ii) $a(ak^2 + bk + c) > 0$, (iii) $k > \frac{-b}{2a}$ must hold true

(A) Statement 1 is true but statement 2 is false

(B) Statement 2 is true but statement 1 is false

(C) Statement 1 is true, statement 2 is true, and statement 2 is a correct explanation of statement 1

(D) Statement 1 is true, statement 2 is true, and statement 2 is a not correct explanation of statement 1

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58. Let $P = 1^2 + (2^2 \times 3^2) + (4^2 \times 5^2) + ... + (24^2 \times 25^2)$,

$Q = (1^2 \times 2^2) + (3^2 \times 4^2) + ... + (23^2 \times 24^2) + 25^2$ and N = (Q – P), then consider the following statements

I) $N > 0$ II) $N < 0$

III) Number of positive integral divisors of $|N| + 624$ is 72

IV) $|N| + 624$ is a perfect cube of an integer

Then number of correct statements

A) 1 B) 3 C) 4 D) 2

59. Let z be a complex number satisfying equation $Z^p = \bar{Z}^q$ where $p, q \in N$, then

I) If p = q, then number of solutions of equation will be infinite

II) If p = q, then number of solutions of equation will be finite

III) If $p \neq q$ then number of solutions of equation will be $p + q + 1$

IV) If $p \neq q$, then number of solutions of equation will be $p + q$

Then number of correct statements are

A) 4 B) 3 C) 2 D) 1

60. Let $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + ... + {}^nC_nx^n$, then

STATEMENT – I: ${}^nC_0 - {}^nC_2 + {}^nC_4 - {}^nC_6 + ... = 2^{\frac{n}{2}}$, where $n = 8k$, where $k \in Z$

STATEMENT – II: ${}^nC_1 - {}^nC_3 + {}^nC_5 - {}^nC_7 + ... = 2^{\frac{n}{2}}$, where $n = 8k + 2$, where $k \in Z$

Then which of the following is true

A) Both STATEMENT – I and STATEMENT – II are true

B) Both STATEMENT – I and STATEMENT – II are false

C) STATEMENT – I is false and STATEMENT – II is true

D) STATEMENT – I is true and STATEMENT – II is false

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61. Let $P\left(\frac{A}{E_i}\right) \propto i$ and $P(E_i) \propto i^3$, for i = 1, 2, 3, and $\sum_{i=1}^{3} P\left(\frac{A}{E_i}\right) = \frac{9}{100}$ and $\sum_{i=1}^{n} P(E_i) = \frac{9}{10}$. Then

(i) $P(A \cap E_2) = \frac{3}{500}$ (ii) $P(A \cap E_1) = \frac{3}{8000}$

(iii) $P\left(\frac{E_2}{A}\right) = \frac{8}{49}$ (iv) $P\left(\frac{E_1}{A}\right) = \frac{1}{98}$

Then, how many of these 4 statements are true?

A) 4 B) 3 C) 2 D) 1

62. If z_1 and z_2 are non-zero complex numbers satisfying $z_1^2 - i\left(\sqrt{2}+1\right)z_1z_2 - \sqrt{2}z_2^2 = 0$, then for the triangle formed by points origin, z_1, z_2 as vertices (read the statements below)

I) Always right-angled triangle

II) Always right-angled isosceles triangle

III) An angle of triangle can be $\tan^{-1}\left(\sqrt{2}\right)$

IV) Area of triangle can be $2\sqrt{2}$ if $|z_1| = 2\sqrt{2}$

Then number of correct statements

A) 1 B) 2 C) 3 D) 4

63. Which of the following statement is INCORRECT?

(A) If the terms of the AP are chosen at regular intervals, then they form an AP

(B) If a constant 'k' is added to each term of AP, with common difference 'd', then the resulting sequence also will be in AP, with common difference 'd'

(C) If n^{th} term of sequence is given as $a_n = \alpha n + \beta$ (linear expression in n) then the sequence is A.P with first term α

(D) If the ratio of nth terms of two A.P's is given then the ratio of the sums of their 'n' terms may be obtained by replacing n with $\frac{n+1}{2}$ in the given ratio

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64. The value of the definite integral $\int_{\pi/8}^{3\pi/8} \frac{11+4\cos 2x+\cos 4x}{1-\cos 4x}dx$ equals

A) $-6-\frac{\pi}{4}$ B) $6\sqrt{2}-\frac{\pi}{4}$ C) $12-\frac{\pi}{2}$ D) $6-\frac{\pi}{4}$

65. **STATEMENT – I:** Let $A=\{0,1,2,3,4,5,6,7\}$, then the number of bijective functions $f:A\to A$ such that $f(1)+f(2)=3-f(3)$ is equal to 720

STATEMENT – II: Let D_n denote the number of ways of derangement of n symbols, then ${}^7C_0.D_7+{}^7C_1.D_6+.....+{}^7C_7.D_0=5040$ (assume $D_0=1$).

Then which of the following is true

A) STATEMENT – I is true and STATEMENT – II is false

B) STATEMENT – I is false and STATEMENT – II is true

C) Both STATEMENT – I and STATEMENT – II are false

D) Both STATEMENT – I and STATEMENT – II are true

66. **STATEMENT – I:** Let 'a' be the arithmetic mean and b, c be the two geometric means between any two positive numbers, then $b^3 + c^3$ equals 2abc

STATEMENT – II: If a, b, c be the pth, q^{th} and r^{th} term, respectively of an AP and GP both, then the product of roots of equation $(a^bb^cc^a)x^2-(abc)x+a^cb^ac^b = 0$, is equal to -1

Then which of the following is true

A) STATEMENT – I is false and STATEMENT – II is true

B) STATEMENT – I is true and STATEMENT – II is false

C) Both STATEMENT – I and STATEMENT – II are false

D) Both STATEMENT – I and STATEMENT – II are true

67. A fair coin is tossed 10 times and the outcomes are listed. Let H_i be the event that the ith outcome is a head and A_m be the event that the list contains exactly m heads, then

A) H_3 and A_4 are independent B) A_1 and H_7 are independent

C) H_2 and A_5 are independent D) A_1 and A_9 are independent

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68. OAB is a sector of a circle of radius R in which O (where O is the centre of the circle), A is fixed point and B is a variable point. Then, the locus of centre of a circle inscribed in the sector OAB is equal to / part of (read the statements below)

I) a parabola with latus rectum 2R

II) a parabola with latus recutm R

III) a parabola with directrix parallel to OA

IV) a parabola with directrix parallel to OB

Then number of correct statements

A) 1 B) 2 C) 3 D) 4

69. Let $f(x) = x^{13} + 2x^{12} + 3x^{11} + ... + 13x + 14$ and

$\alpha = \cos\frac{2\pi}{15} + i\sin\frac{2\pi}{15}, \left(i = \sqrt{-1}\right)$. If $N = f(\alpha).f(\alpha^2).f(\alpha^3).....f(\alpha^{14})$ then (read the statements below)

I) number of postive intgral divisors of N is 196

II) number of positive integral divisors of N is256

III) number of positive integral divisors of n which are perfect squares is 49

IV) number of positive integral divisors of n which are perfect cubes is 9

Then number of correct statements

A) 2 B) 3 C) 1 D) 4

70. Order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^4 + \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^5 + \sin\left(\frac{dy}{dx}\right) = 1$, respectively is

(A) 2, 4 (B) 1, 5 (C) 2, 1 (D) 2, not defined

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. Let $I=\int\frac{\left(e^x-1\right)\left(\sin x-\cos x\right)+x\cos x}{\sin^2 x+\left(e^x-1-x\right)^2}dx=\text{Tan}^{-1}\left(f\left(x\right)\right)+C$, where 'C' is the constant of integration and $\lim\limits_{x\to0}\frac{f\left(x\right)}{x}=\frac{p}{q}$ where $p,q\in N$ then find the least value of p + q.

72. How many ordered pair of integers (a, b) satisfy all the following qualities.

$a^2+b^2<16,\ a^2+b^2<8a,\ a^2+b^2<8b$

73. Total number of solutions of the equation $x + 3y + 4z = 40$, $x, y, z \in W$, is

74. Let $(1+x)^n={}^nC_0+{}^nC_1x+{}^nC_2x^2+.....+{}^nC_nx^n$ and $\int\limits_0^a x^n\,dx=\frac{a^{n+1}}{n+1}$. If

$${}^{20}C_0+{}^{20}C_1+\frac{4}{3}\cdot{}^{20}C_2+.....=\frac{\alpha^{21}-1}{\beta},$$ then the value of $\alpha+\beta$, is _____.

75. For the hyperbola $\frac{(x-2)^2}{9}-\frac{\left(y+2\right)^2}{16}=1$, consider the following statements

(i) Center: (2, –2)

(ii) eccentricity = $\frac{5}{4}$

(iii) Length of latus rectum = $\frac{32}{3}$

(iv) Equation of one of the directrix is $x=\frac{19}{5}$

(v) Equation of its conjugate hyperbola is $\frac{(y+2)^2}{16}-\frac{(x-2)^2}{9}=1$

(vi) Point (12, 8) lie outside of the given hyperbola

(vii) Equation of director circle is $x^2+y^2=7$

Then, how many of these 7 statements are true?

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